Dr. Hidetaka Kuniyoshi (he/him)

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Current Position

1. Japan Society for the Promotion of Science (JSPS) Overseas Research Fellow (Apr. 2025-)
   Host institution: Solar and Space Physics Group, Northumbria University, UK

Education

1. **PhD** (awarded 24th March 2025)
   Dpt. of Earth and Planetary Science, The University of Tokyo (Apr. 2022–Mar. 2025)
   Supervisors: Prof. Takaaki Yokoyama & Prof. Shinsuke Imada
   Thesis*:* The role of swirls in solar coronal heating
2. **MSc** (awarded 24th March 2022)
   Dpt. of Earth and Planetary Science, The University of Tokyo (Apr. 2020–Mar. 2022)
3. **BSc** (awarded 24th March 2020)
   Dpt. of Earth and Planetary Physics, The University of Tokyo (Apr. 2016–Mar. 2020)

Fellowships, Awards & Grants

1. Japan Society for the Promotion of Science (JSPS) Overseas Postdoctoral Research Fellowships, April 2025-March2027
2. International Astronomical Union (IAU) PhD Prize 2025
3. The University of Tokyo School of Science Research Award (Doctor, ranked 1st in Dpt. of Earth and Planetary Science), March 2025
4. The University of Tokyo School of Science Research Award (Master), March 2022
5. International Travel Support for Students, Institute for Space-Earth Environmental Research (ISEE), Nagoya University, June 2023 (JPY 450,000)
6. Bronze Medal, Japan Physics Olympiad, August 2014

Publications (full list in NASA ADS)

Manuscript under Review

1. **Kuniyoshi, H.**, & Morton, R. J., 2026, *A self-consistent solar coronal heating model by Alfvénic waves*, submitted to *ApJ*.

Refereed Papers as Corresponding Author

2. **Kuniyoshi, H.**, Imada, S., Yokoyama, T., 2025, *A unified picture of swirl-driven coronal heating: magnetic energy supply and dissipation*, *ApJL*, 990, L71. (NASA/ADS)
3. **Kuniyoshi, H.**, Bose, S., Yokoyama, T., 2024, *Comprehensive Synthesis of Magnetic Tornado: Cospatial Incidence of Chromospheric Swirls and Extreme-ultraviolet Brightening*, *ApJL*, 969, L34. (NASA/ADS)

4. **Kuniyoshi, H.**, Morton, R. J., Shoda, M., Yokoyama, T., 2024, *Can the Solar p-modes Contribute to the High-frequency Transverse Oscillations of Spicules?*, *ApJ*, 960, 118. (NASA/ADS).

5. **Kuniyoshi, H.**, Shoda, M., Iijima, H., Yokoyama, T., 2023, *Magnetic Tornado Properties: A Substantial Contribution to the Solar Coronal Heating via Efficient Energy Transfer*, *ApJ*, 949, 8. (NASA/ADS)

6. **Kuniyoshi, H.**, Hesse, M., Norgren, C., Tenfjord, P., Kwagala, N. K., 2021, *Magnetic Reconnection in a Sheared Magnetic Flux Tube: Slippage Versus Tearing*, *JGR: Space Physics*, 126, e29236. (NASA/ADS)

Other Refereed Paper

7. Baker, D., et al. (incl. **Kuniyoshi, H.**), 2024, *Searching for Evidence of Subchromospheric Magnetic Reconnection on the Sun*, ApJ, 970, 39. 6 times cited (NASA/ADS)

Non-refereed Paper

8. Silva, S. S. A., et al. (incl. **Kuniyoshi, H.**), *Observing solar vortices with existing and future instrumentation. Solar Physics International Network for Swirls (SPINS) white paper (Helio)*, White paper submitted to UK Space Frontiers 2035 on 28 (NASA/ADS)

High Performance Computer (HPC) and Observational experience as PI

HPC

1. **ATERUI III**, *12 million CPU hours*, Centre for Computational Astrophysics (CfCA) in National Astronomical Observatory of Japan (NAOJ), Japan, 2026
2. **ATERUI III**, *12 million CPU hours*, CfCA in NAOJ, Japan, 2025
3. **ATERUI II**, *12 million CPU hours*, CfCA in NAOJ, Japan, 2024
4. **DiRAC**, *2.8 million CPU hours*, Science and Technology Facilities Council (STFC), UK, 2026
5. **A-KDK**, *1.7 million CPU hours*, Research Institute for Sustainable Humanosphere (RISH) in Kyoto Univ., Japan, 2026
6. **A-KDK**, *1.7 million CPU hours*, RISH in Kyoto Univ., Japan, 2025
7. **A-KDK**, *1.7 million CPU hours*, RISH in Kyoto Univ., Japan, 2024

Observation

8. **Daniel K. Inouye Solar Telescope (DKIST)**, *3.2 hours,* DKIST Cycle 4, 2026

Oral Presentations at Conferences (3 invited international talks; 2 invited domestic talks)

2026 (1 invited international talks; 1 invited domestic talk)

9. Summer School for Young Astronomers in Japan, "Current situation and frontiers of the solar coronal heating problem", Nagano, Japan, 30 Jul. 2026
10. National Astronomical Meeting (NAM), "A self-consistent solar coronal heating model by Alfvenic waves", Birmingham, UK, 20–24 Jul. 2026
11. Coronal Loop Workshop XII, "A self-consistent solar coronal heating model by Alfvenic waves", Newcastle, UK, 10 Jun. 2026
12. Kyoto University A-KDK Supercomputer Symposium, "A self-consistent solar coronal heating model by Alfvenic waves", Online, 05 Mar. 2026

13. International Conference on “Exploring the Sun at High Resolution: Present Perspectives & Future Horizons”, “A unified picture of swirl-driven coronal heating: magnetic energy supply and dissipation.”, Udaipur, India, 10-13 Feb. 2026 (**invited**)
14. Center for Computational Astrophysics (CfCA) Users Meeting, “A self-consistent solar coronal heating model by Alfvenic waves”, Tokyo, Japan, 23 Jan. 2026

2025 (2 invited international talks; 1 invited domestic talk)

15. Rironkon Symposium, “A self-consistent solar coronal heating model by Alfvenic waves”, Ibaraki, Japan, 17–19 Dec. 2025
16. Waves and Instability in the Solar Atmosphere (WISA), “A unified picture of swirl-driven coronal heating: magnetic energy supply and dissipation.”, Rome, Italy, 23-26 Sep. 2025
17. Hinode18–IRIS16 Meeting, “A unified picture of swirl-driven coronal heating: magnetic energy supply and dissipation.”, London, UK, 23–27 Jun. 2025 (**invited**)
18. Solar Physics International Network for Swirls (SPINS) Meeting, “A unified picture of swirl-driven coronal heating: magnetic energy supply and dissipation.”, Sheffield, UK, 2–4 Jun. 2025 (**invited**)
19. Astronomical Society of Japan (ASJ) Spring Meeting, “A unified picture of swirl-driven coronal heating: magnetic energy supply and dissipation.”, Ibaraki, Japan, 17–20 Mar. 2025
20. Japan Solar Physics Community (JSPC) Symposium, “A unified picture of swirl-driven coronal heating: magnetic energy supply and dissipation.”, Kanagawa, Japan, 17–19 Feb. 2025 (**invited**)
21. Cool Stars Workshop in Japan, “A unified picture of swirl-driven coronal heating: magnetic energy supply and dissipation.”, Tokyo, Japan, 12–14 Feb. 2025

2024 (1 invited domestic talk)

22. Rironkon Symposium, “A unified picture of swirl-driven coronal heating: magnetic energy supply and dissipation.”, Tokyo, Japan, 24–26 Dec. 2024
23. ASJ Fall Meeting, “Comprehensive Synthesis of Magnetic Tornado: Cospatial Incidence of Chromospheric Swirls and Extreme-ultraviolet Brightening”, Hyogo, Japan, 11–13 Sep. 2024
24. Hinode-17 IRIS-15 SPHERE-3 Meeting, “Comprehensive Synthesis of Magnetic Tornado: Cospatial Incidence of Chromospheric Swirls and Extreme-ultraviolet Brightening”, Montana, USA, 23–27 Jul. 2024
25. Research Institute for Mathematical Sciences (RIMS) Symposium “Non-equilibrium Turbulence”, “Comprehensive Synthesis of Magnetic Tornado: Cospatial Incidence of Chromospheric Swirls and Extreme-ultraviolet Brightening”, Kyoto, Japan, 22–24 Jul. 2024 (**invited**)
26. ASJ Spring Meeting, “Quiet Sun small-scale brightening initiated by a magnetic tornado”, Tokyo, Japan, 11–15 Mar. 2024
27. Solar-C SWG Meeting, “Quiet Sun small-scale brightening initiated by a magnetic tornado”, Nagoya, Japan, 5–6 Mar. 2024
28. JSPC Symposium, “Quiet Sun small-scale brightening initiated by a magnetic tornado”, Tokyo, Japan, 20–22 Feb. 2024
29. CfCA Users Meeting, “Quiet Sun small-scale brightening initiated by a magnetic tornado”, Tokyo, Japan, 29–30 Jan. 2024

2023

30. Hinode-16 / IRIS-13 Meeting, “Can the Solar p-modes Contribute to the High-frequency Transverse Oscillations of Spicules?”, Niigata, Japan, Sep. 2023
31. ASJ Fall Meeting, Nagoya, “Can the Solar p-modes Contribute to the High-frequency Transverse Oscillations of Spicules?”, Japan, Sep. 2023

32. WISA 2023, ”Magnetic Tornado Properties: A Substantial Contribution to the Solar Coronal Heating via Efficient Energy Transfer”, Newcastle, UK, Jun. 2023
33. ASJ Spring Meeting, ”Magnetic Tornado Properties: A Substantial Contribution to the Solar Coronal Heating via Efficient Energy Transfer”, Tokyo, Japan, Mar. 2023
34. JSPC Symposium, ”Magnetic Tornado Properties: A Substantial Contribution to the Solar Coronal Heating via Efficient Energy Transfer”, Nagoya, Japan, Feb. 2023

2022

35. ASJ Fall Meeting, ”MBP merger, an alternative solar/stellar coronal heating mechanism” Niigata, Japan, Sep. 2022
36. Japan Society of Fluid Mechanics (JSFM) Annual Meeting, ”MBP merger, an alternative solar/stellar coronal heating mechanism”, Kyoto, Japan, Sep. 2022
37. SOLAR MHD Conference, ”MBP merger, an alternative solar/stellar coronal heating mechanism”, Eastbourne, UK, Aug. 2022
38. Japan Geophysical Union (JpGU) Meeting, ”MBP merger, an alternative solar/stellar coronal heating mechanism”, Chiba, Japan (hybrid), May 2022
39. JSPC Symposium, ”MBP merger, an alternative solar/stellar coronal heating mechanism”, Online, Feb. 2022

2021

40. American Geophysical Union (AGU) Fall Meeting, ”MBP merger, an alternative solar/stellar coronal heating mechanism”, New Orleans, USA (hybrid), Dec. 2021
41. JpGU Meeting, “Magnetic Reconnection in a Sheared Magnetic Flux Tube: Slippage Versus Tearing”, Online, May 2021
42. ASJ Spring Meeting, “Magnetic Reconnection in a Sheared Magnetic Flux Tube: Slippage Versus Tearing”, Online, Mar. 2021

Poster presentations at Conferences and Workshops

2026

1. Cool Stars 23, Ariake, “A self-consistent solar coronal heating model by Alfvenic waves”, Japan, 15-16 Jun. 2026

2025

2. International Astronomical Union (IAU) Symposium 400, “A unified picture of swirl-driven coronal heating: magnetic energy supply and dissipation.”, Medellín, Colombia, 21–25 Jul. 2025
3. National Astronomical Meeting (NAM), “A unified picture of swirl-driven coronal heating: magnetic energy supply and dissipation.”, Durham, UK, 7–11 Jul. 2025

2024

4. CfCA Users Meeting, “A unified picture of swirl-driven coronal heating: magnetic energy supply and dissipation.”, Tokyo, Japan, 25–26 Nov. 2024

2023

5. Rironkon Symposium, “Quiet Sun small-scale brightening initiated by a magnetic tornado”, Aomori, Japan, Dec. 2023
6. AGU Fall Meeting, ”Magnetic Tornado Properties: A Substantial Contribution to the Solar Coronal Heating via Efficient Energy Transfer”, San Francisco, USA, Dec. 2023

7. JpGU Meeting, “Can the Solar p-modes Contribute to the High-frequency Transverse Oscillations of Spicules?”, Chiba, Japan, May 2023

2022

8. IAU General Assembly, ”MBP merger, an alternative solar/stellar coronal heating mechanism”, Busan, ROK (hybrid), Aug. 2022

2020

9. AGU Fall Meeting, “Magnetic Reconnection in a Sheared Magnetic Flux Tube: Slippage Versus Tearing”, Online, Dec. 2020
10. JSPC Symposium, “Magnetic Reconnection in a Sheared Magnetic Flux Tube: Slippage Versus Tearing”, Online, Dec. 2020

Invited Seminar Talks

1. Jim Klimchuk Lab Seminar, NASA Goddard Space Flight Center, online, Sep. 2026
2. IAU Early Career Summer Talk Series, online, Aug. 2026
3. Plasma Dynamics Group (PDG) seminar, Sheffield University, UK, Oct. 2025
4. Solar Orbiter Atmospheric Heating WG Seminar, online, May 2025
5. Yamada Lab Seminar, Waseda University, Japan, May 2025
6. ISEE Seminar, Nagoya University, Aichi, Japan, Jul. 2024
7. Tsukuba Space Forum, University of Tsukuba, Japan, Apr. 2024
8. Chromospheric Magnetic Field Workshop, National Astronomical Observatory, Japan, Feb. 2024
9. Dept. of Earth Science & Astronomy Seminar, University of Tokyo, Japan, Oct. 2023
10. Mullard Space Science Laboratory Seminar, University College London, Aug. 2023
11. Plasma Dynamics Group (PDG) seminar, Sheffield University, UK, May 2023

Conference organizing experiences

1. **Session chair**, “Linking Plasma Dynamics in the Lower Solar Atmosphere to Magnetic Activity”, Asia Oceania Geoscience Society, Fukuoka Japan, Aug. 2026 (accepted),
2. **Session co-chair**, “Symphony of Waves and Plasma (SWAP): Insights into Solar Atmospheric Coupling in the High-Resolution Era”, National Astronomical Meeting (NAM), Birmingham UK, Jul. 2026
3. **LOC**, Coronal Loop Workshop XII, Newcastle UK, Jun. 2026